

2018-07-15 TRAFFIC ANALYSIS EXERCISE: ANSWERS

OH NOES! TORRENTZ ON OUR NETWORK!

www.malware-traffic-analysis.net/2018/06/30/2018-07-15-traffic-analysis-exercise.pcap.zip

Zip files are password-protected with the standard password. See the "about" page of my site if you don't know it.

The associated network segment:

- LAN segment: 10.0.0.0/24 (10.0.0.0 through 10.0.0.255)
- Broadcast address: 10.0.0.255
- Domain controller: 10.0.0.2 (DogOfTheYear-DC)
- Domain: dogoftheyear.net

SCENARIO:

You have received alerts on bittorrent traffic from 10.0.0.201 on your organization's network. Torrent traffic is often associated with file sharing of copyright-protected content; however, many cases of torrent traffic are perfectly legal (like this traffic analysis exercise).

QUESTIONS:

Based on the pcap, answer the following questions:

- What is the MAC address of the computer at 10.0.0.201?
- What is the host name of the computer at 10.0.0.201?
- What is the Windows user account name for the computer at 10.0.0.201?
- What is the Microsoft Windows version (XP, 7, 8, or 10) of the computer at 10.0.0.201?
- What time in UTC did the torrent activity from 10.0.0.201 start?
- What torrent file did the user at 10.0.0.201 download?
- What is the name of the torrent client used on 10.0.0.201?
- What file is being seeded (shared) by the torrent client on 10.0.0.201?

ANSWERS

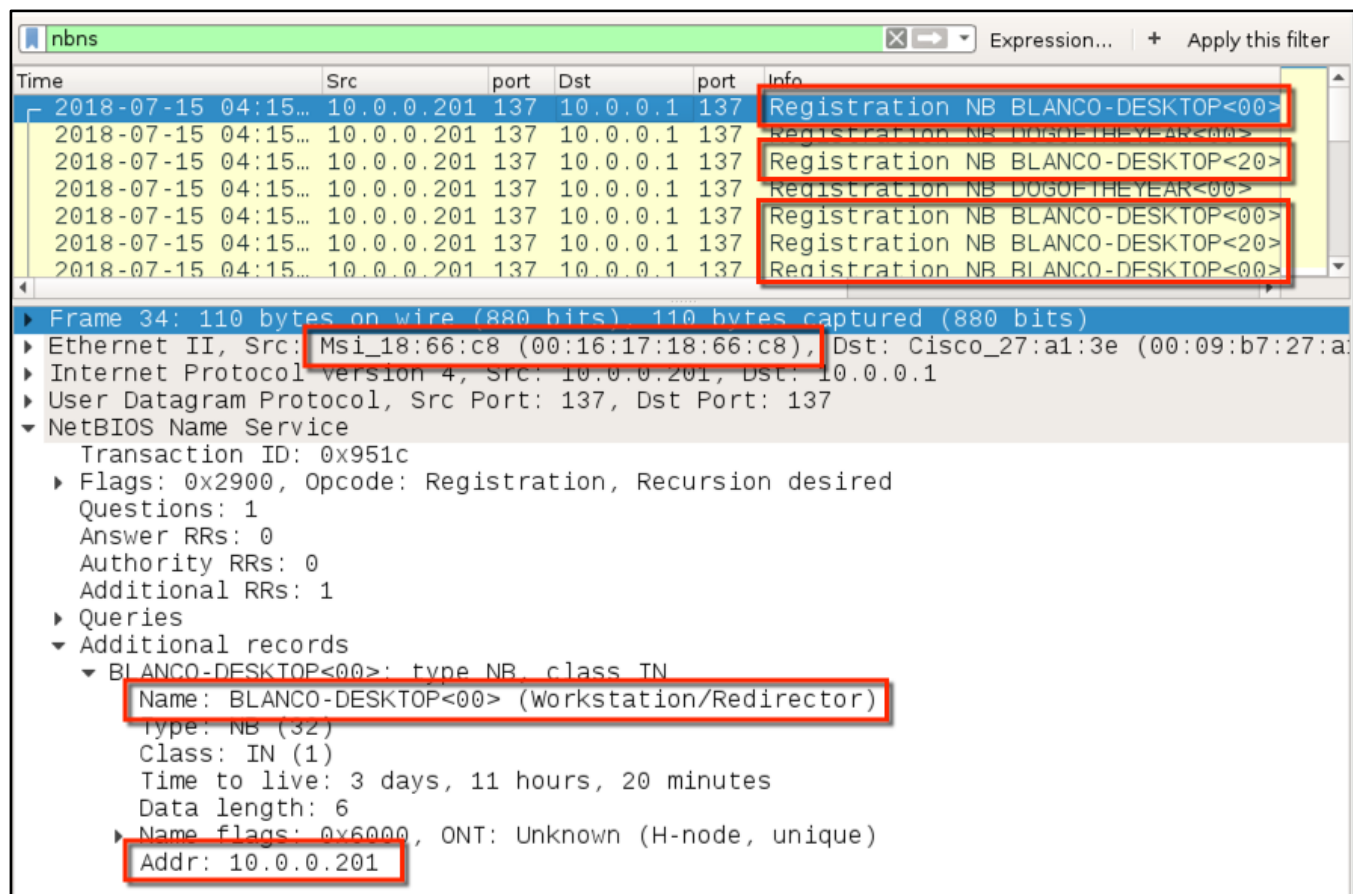
- IP address: 10.0.0.201
- MAC address: 00:16:17:18:66:c8 (Msi_18:66:c8)
- Host name: BLANCO-DESKTOP
- Windows user account name: elmer.blanco
- Windows version: Windows 10
- Date/time of torrent activity started: Sunday 2018-07-15 at 04:17 UTC
- Torrent file downloaded by 10.0.0.201: Betty_Boop_Rhythm_on_the_Reservation.avi.torrent
- Torrent client: Deluge version 1.3.15
- Torrent file being seeded (shared): ubuntu-18.04-desktop-amd64.iso (SHA1 info hash e4be9e4db876e3e3179778b03e906297be5c8dbe)

2018-07-15 TRAFFIC ANALYSIS EXERCISE: ANSWERS

DETAILS:

As always, I recommend you set up Wireshark by changing the default column display according to the tutorials I've posted [here](#).

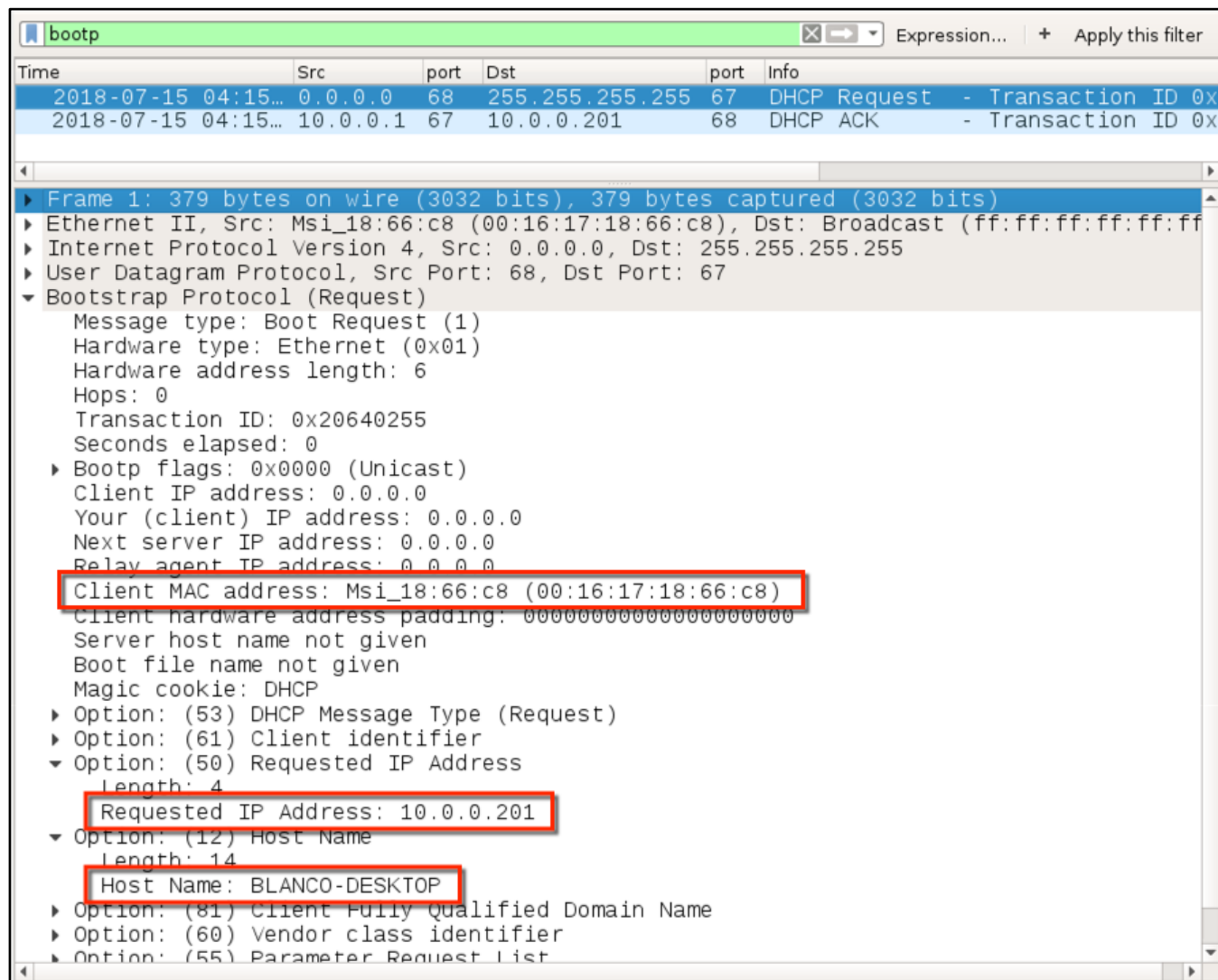
Finding the host and user information is the same as we've done in previous exercises. You can easily correlate the IP address of a Windows host with its host name and MAC address by searching on the NBNS traffic (Wireshark filter: **nbns**) as shown in the image below.



Shown above: Correlating the IP address, MAC address, and host name based on NBNS traffic.

2018-07-15 TRAFFIC ANALYSIS EXERCISE: ANSWERS

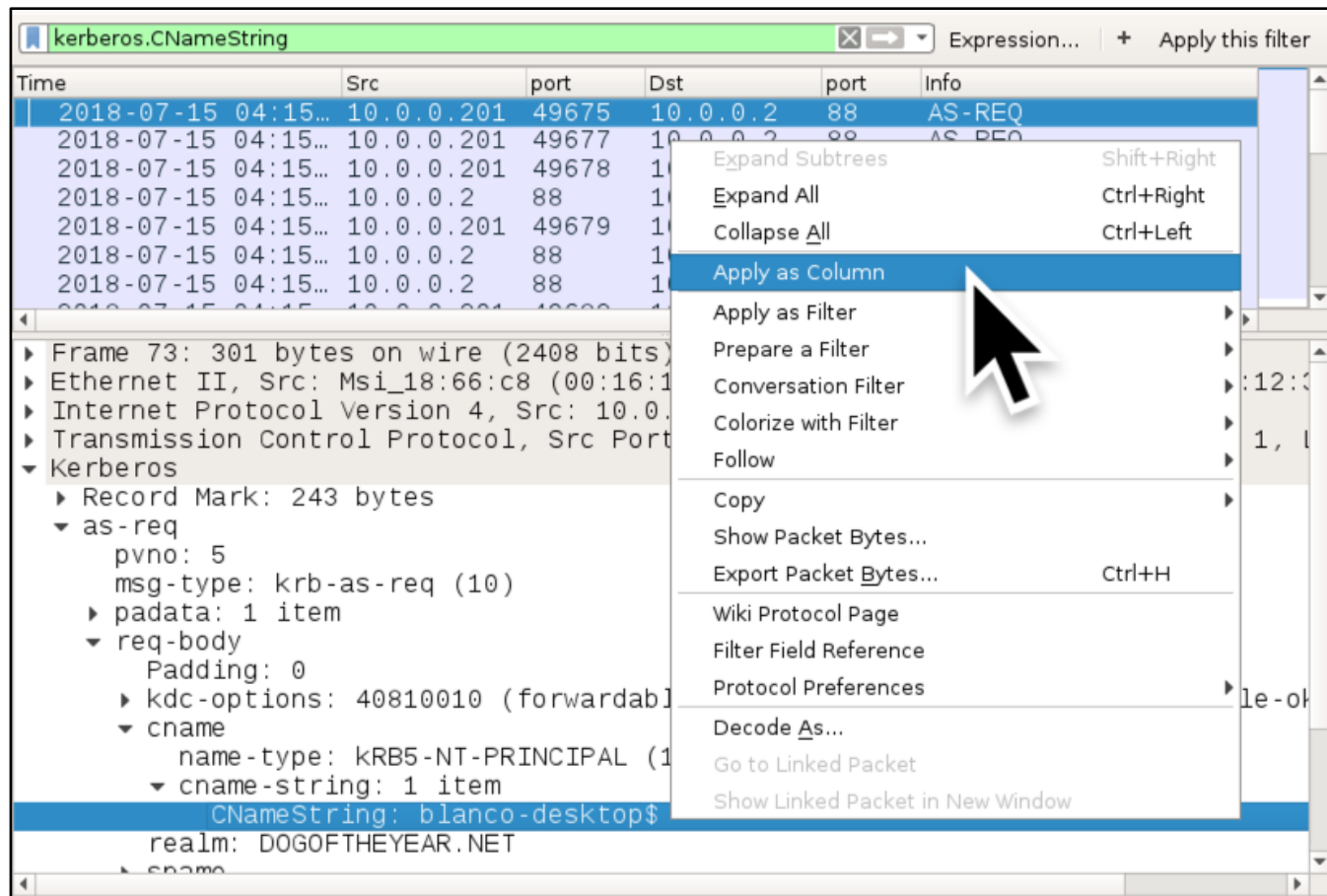
You can also correlate the IP address with the MAC address and host name from the DHCP traffic (Wireshark filter: **bootp**) as shown in the next image.



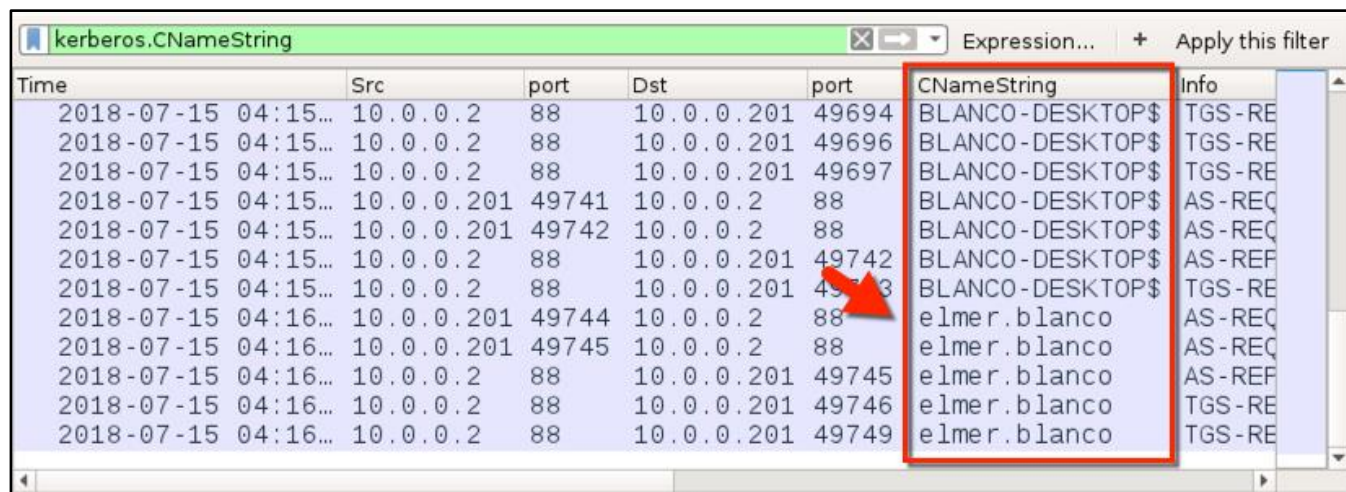
Shown above: Finding the hostname, IP address, and MAC address based on DHCP traffic.

2018-07-15 TRAFFIC ANALYSIS EXERCISE: ANSWERS

Search Kerberos to find the Windows user account name (Wireshark filter: **kerberos.CNameString**). CNameString values represent Windows host names and user account names. Once you locate any CNameString value in the frame details panel, apply it as a column, so you can quickly find the associated user account from your column display.



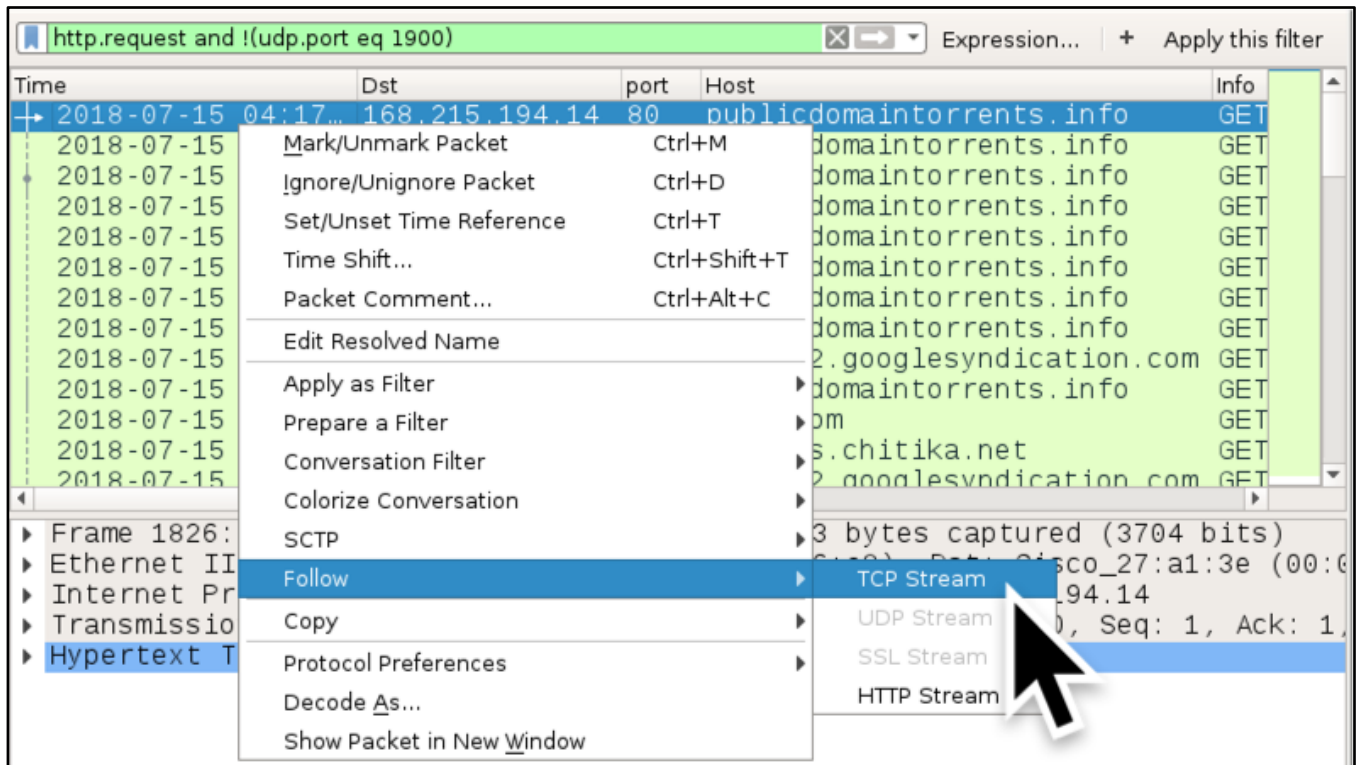
Shown above: Applying one of the CNameString values as a column.



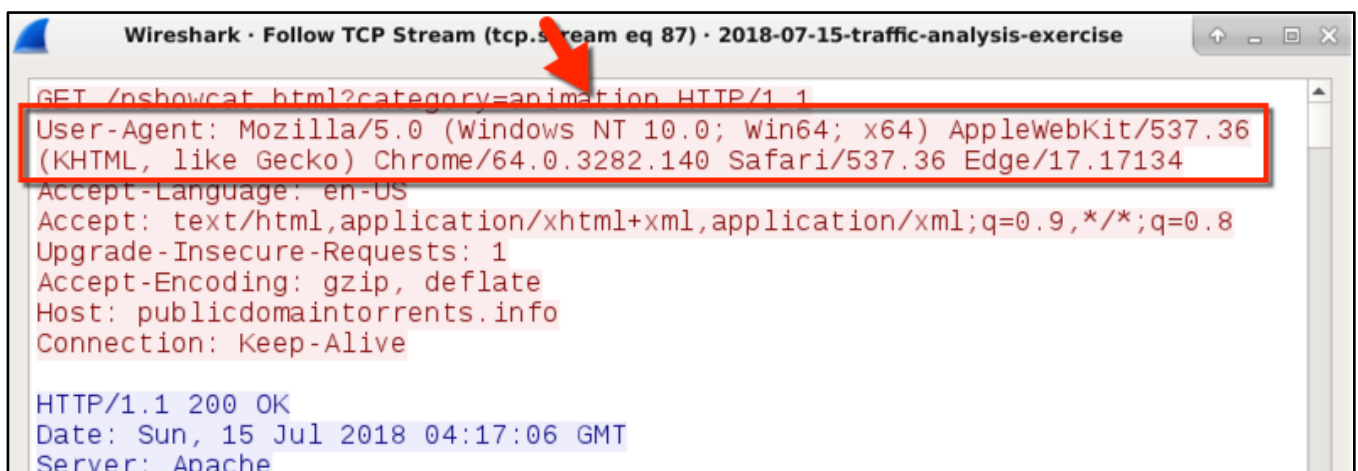
Shown above: CNameString value as a column. All host names end with \$. User account names are ones without the \$ sign at the end.

2018-07-15 TRAFFIC ANALYSIS EXERCISE: ANSWERS

How do we find which version of Windows this host is running? Check the User-Agent string in the HTTP request headers for any normal web browsing traffic. In this pcap, we can filter on ***http.request and !(udp.port eq 1900)*** to see the user viewed ***publicdomaintorrents.info***. Follow the TCP stream for the first HTTP GET request and check the User-Agent string. You'll find this is a Windows 10 host.



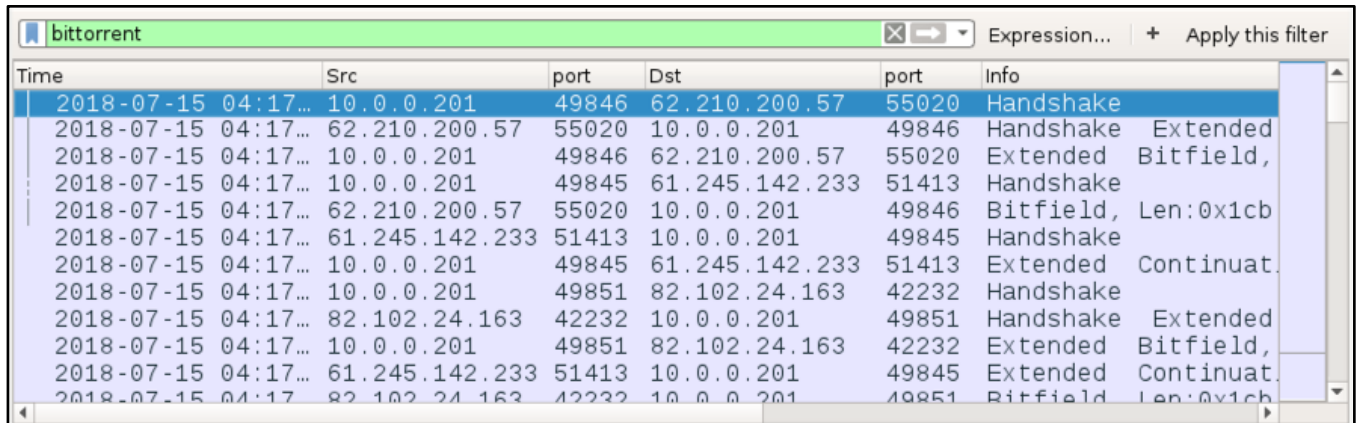
Shown above: Following the TCP stream for the first HTTP GET request.



Shown above: The User-Agent string reveals this is a 64-bit Windows 10 host.

2018-07-15 TRAFFIC ANALYSIS EXERCISE: ANSWERS

When did the torrent traffic start? Filter on **bittorrent** and you'll see 2018-07-15 at 04:17 UTC.

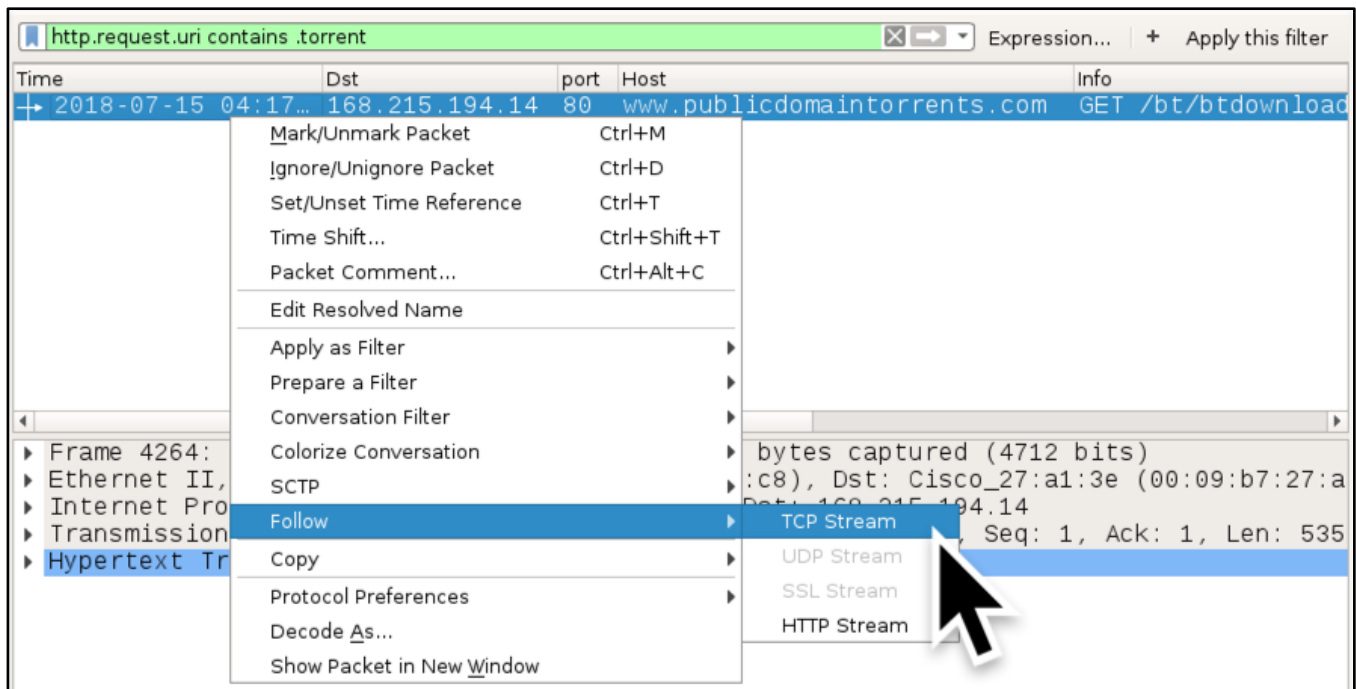


The screenshot shows the Wireshark interface with a filter 'bittorrent' applied. The packet list table is as follows:

Time	Src	port	Dst	port	Info
2018-07-15 04:17...	10.0.0.201	49846	62.210.200.57	55020	Handshake
2018-07-15 04:17...	62.210.200.57	55020	10.0.0.201	49846	Handshake Extended
2018-07-15 04:17...	10.0.0.201	49846	62.210.200.57	55020	Extended Bitfield,
2018-07-15 04:17...	10.0.0.201	49845	61.245.142.233	51413	Handshake
2018-07-15 04:17...	62.210.200.57	55020	10.0.0.201	49846	Bitfield, Len:0x1cb
2018-07-15 04:17...	61.245.142.233	51413	10.0.0.201	49845	Handshake
2018-07-15 04:17...	10.0.0.201	49845	61.245.142.233	51413	Extended Continuat.
2018-07-15 04:17...	10.0.0.201	49851	82.102.24.163	42232	Handshake
2018-07-15 04:17...	82.102.24.163	42232	10.0.0.201	49851	Handshake Extended
2018-07-15 04:17...	10.0.0.201	49851	82.102.24.163	42232	Extended Bitfield,
2018-07-15 04:17...	61.245.142.233	51413	10.0.0.201	49845	Extended Continuat.
2018-07-15 04:17...	82.102.24.163	42232	10.0.0.201	49851	Bitfield Len:0x1cb

Shown above: Filtering on **bittorrent** in Wireshark.

Files with the file extension **.torrent** contain information that guides torrent clients to retrieve a file using the bittorrent protocol. What **.torrent** file did the user at 10.0.0.201 download? To find out, filter for any HTTP GET requests with **.torrent** in the URL. So filter on **http.request.uri contains .torrent** and you should see only one HTTP request. Follow the TCP stream for that result single result.



The screenshot shows the Wireshark interface with a filter 'http.request.uri contains .torrent' applied. The packet list table is as follows:

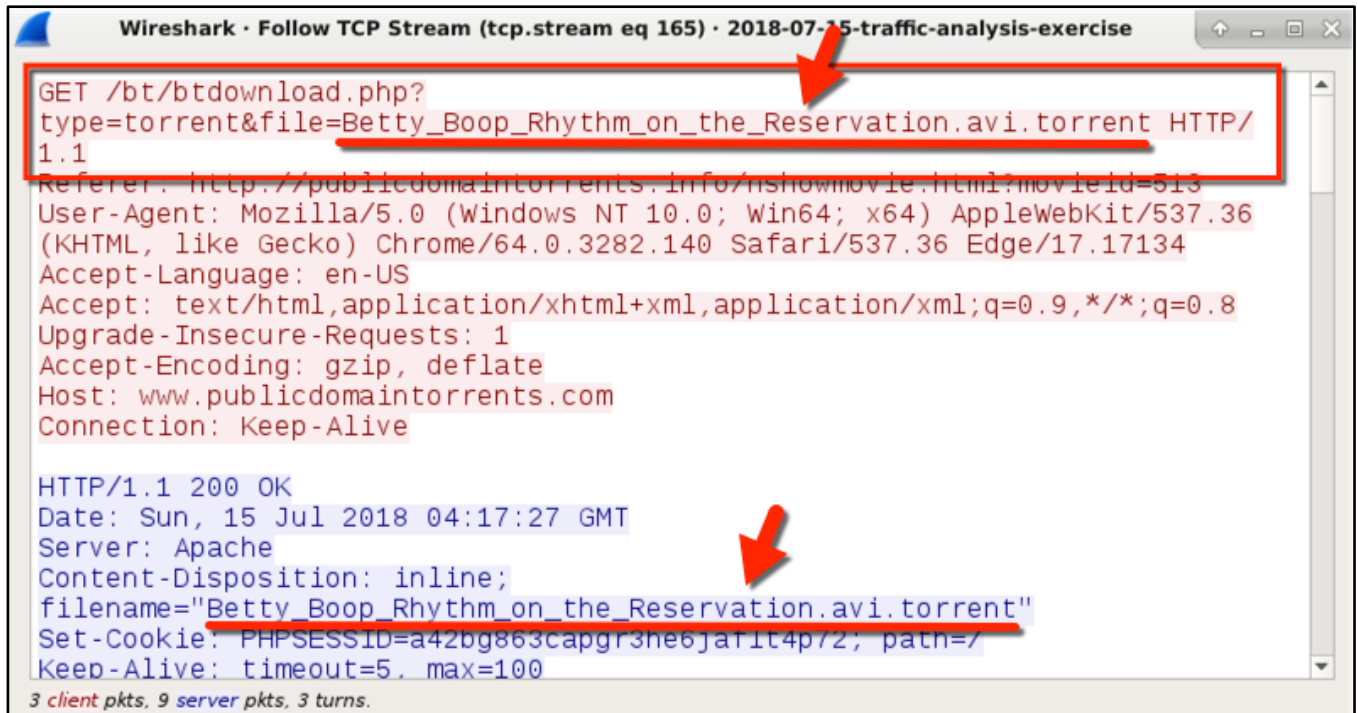
Time	Dst	port	Host	Info
2018-07-15 04:17...	168.215.194.14	80	www.publicdomaintorrents.com	GET /bt/btdownload

A context menu is open over the selected packet, with the 'Follow' option highlighted. The 'Follow' submenu is also open, showing 'TCP Stream' as the selected option. The packet details pane on the right shows the following information:

- bytes captured (4712 bits)
- :c8), Dst: Cisco_27:a1:3e (00:09:b7:27:a
- Seq: 1, Ack: 1, Len: 535

Shown above: One method of finding a **.torrent** file in an HTTP request in Wireshark.

2018-07-15 TRAFFIC ANALYSIS EXERCISE: ANSWERS



```
Wireshark · Follow TCP Stream (tcp.stream eq 165) · 2018-07-15-traffic-analysis-exercise

GET /bt/btdownload.php?
type=torrent&file=Betty_Boop_Rhythm_on_the_Reservation.avi.torrent HTTP/
1.1
Referer: http://publicdomaintorrents.info/nshowmovie.html?movieid=813
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36
(KHTML, like Gecko) Chrome/64.0.3282.140 Safari/537.36 Edge/17.17134
Accept-Language: en-US
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8
Upgrade-Insecure-Requests: 1
Accept-Encoding: gzip, deflate
Host: www.publicdomaintorrents.com
Connection: Keep-Alive

HTTP/1.1 200 OK
Date: Sun, 15 Jul 2018 04:17:27 GMT
Server: Apache
Content-Disposition: inline;
filename="Betty_Boop_Rhythm_on_the_Reservation.avi.torrent"
Set-Cookie: PHPSESSID=a42bg863capgr3he6ja1t4p72; path=/
Keep-Alive: timeout=5, max=100

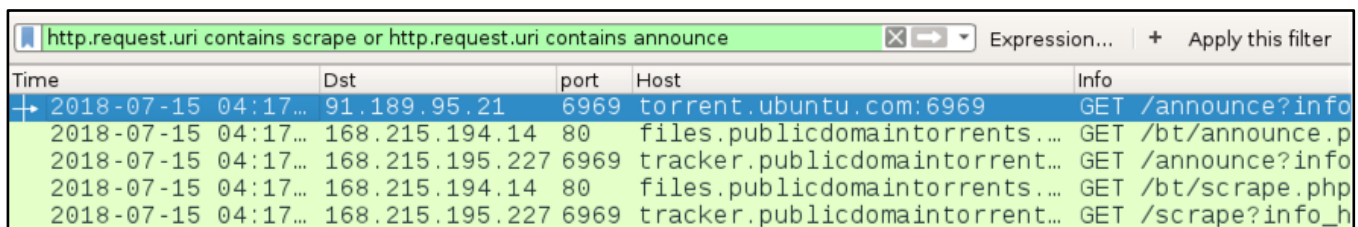
3 client pkts, 9 server pkts, 3 turns.
```

Shown above: Finding a **.torrent** file name from HTTP headers in the TCP stream.

The **.torrent** file is named **Betty_Boop_Rhythm_on_the_Reservation.avi.torrent**.

What about the other torrent traffic? We can search on "announce" or "scrape" in the HTTP URLs to find torrent tracker traffic and see if any other files are being torrented. The search filter in this case is: **http.request.uri contains announce or http.request.uri contains scrape**

HTTP request headers from tracker traffic should also give us the torrent client in the User-Agent string.



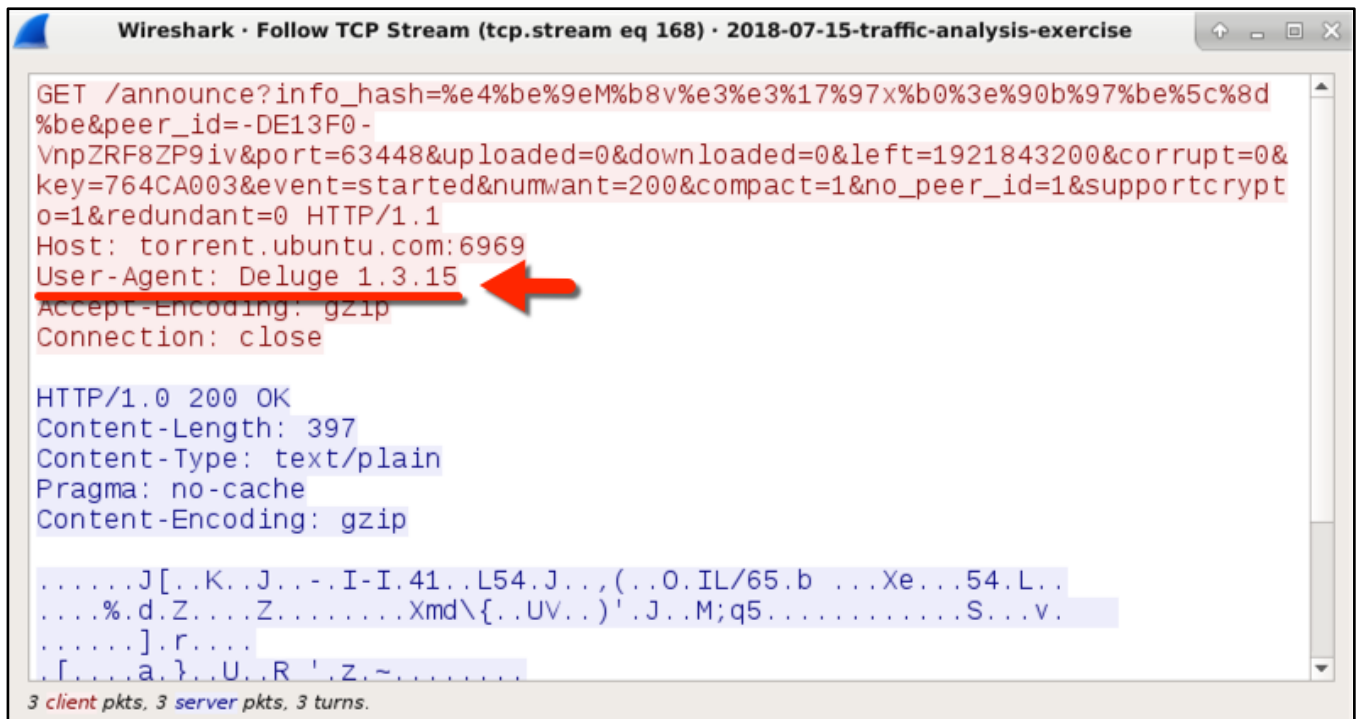
Time	Dst	port	Host	Info
2018-07-15 04:17...	91.189.95.21	6969	torrent.ubuntu.com:6969	GET /announce?info
2018-07-15 04:17...	168.215.194.14	80	files.publicdomaintorrents...	GET /bt/announce.p
2018-07-15 04:17...	168.215.195.227	6969	tracker.publicdomaintorrent...	GET /announce?info
2018-07-15 04:17...	168.215.194.14	80	files.publicdomaintorrents...	GET /bt/scrape.php
2018-07-15 04:17...	168.215.195.227	6969	tracker.publicdomaintorrent...	GET /scrape?info_h

Shown above: Filtering in Wireshark to find HTTP-based torrent tracker traffic.

In addition to tracker traffic to two **publicdomaintorrents.com** domains, the results also show tracker traffic to **torrent.ubuntu.com**.

Follow the TCP stream for the first HTTP request, the one to **torrent.ubuntu.com**. Look at the User-Agent string, and you'll see Deluge 1.3.15. [Deluge](#) is a torrent client with versions for Windows, macOS, Linux, and BSD. It's currently at version 1.3.15 as I write this.

2018-07-15 TRAFFIC ANALYSIS EXERCISE: ANSWERS



```
Wireshark · Follow TCP Stream (tcp.stream eq 168) · 2018-07-15-traffic-analysis-exercise

GET /announce?info_hash=%e4%be%9eM%b8v%e3%e3%17%97x%b0%3e%90b%97%be%5c%8d
%be&peer_id=-DE13F0-
VnpZRF8ZP9iv&port=63448&uploaded=0&downloaded=0&left=1921843200&corrupt=0&
key=764CA003&event=started&numwant=200&compact=1&no_peer_id=1&supportcrypt
o=1&redundant=0 HTTP/1.1
Host: torrent.ubuntu.com:6969
User-Agent: Deluge 1.3.15
Accept-Encoding: gzip
Connection: close

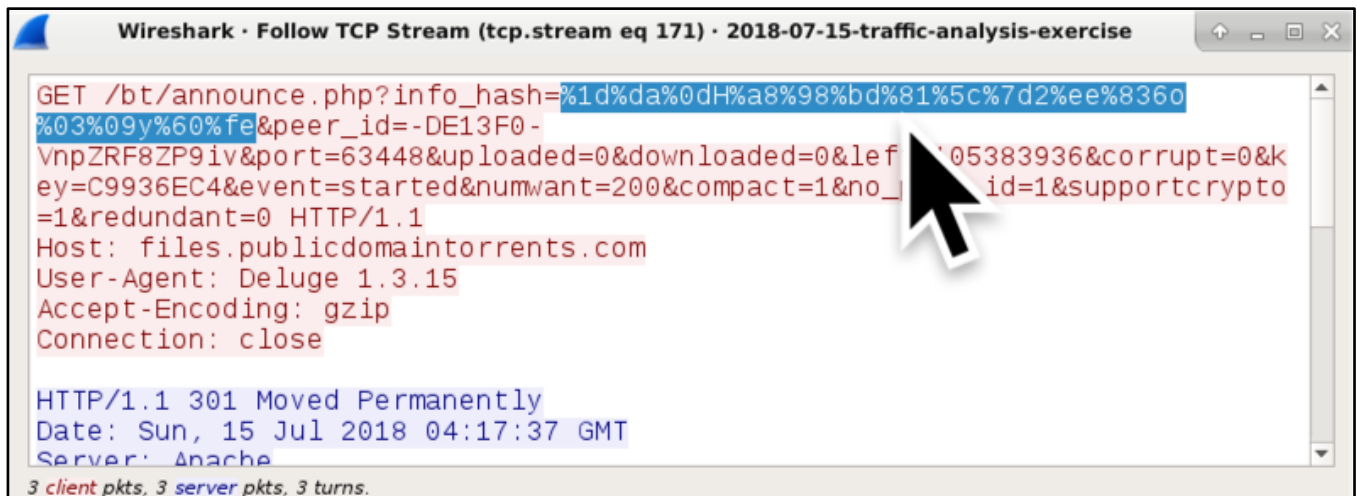
HTTP/1.0 200 OK
Content-Length: 397
Content-Type: text/plain
Pragma: no-cache
Content-Encoding: gzip

.....J[.K...J...-I-I.41...L54.J...(.O.IL/65.b...Xe...54.L..
....%.d.Z...Z.....Xmd\{..UV..)'J..M;q5.....S...V.
.....].r....
[.....a}..U..R'.z~.....

3 client pkts, 3 server pkts, 3 turns.
```

Shown above: User-Agent string showing Deluge as the torrent client.

What is this traffic actually tracking? You can find that out from the **info_hash** value. In the same HTTP GET request, select the value right after **info_hash=** right up to **&peer_id=** as shown in the image below.



```
Wireshark · Follow TCP Stream (tcp.stream eq 171) · 2018-07-15-traffic-analysis-exercise

GET /bt/announce.php?info_hash=%1d%da%0dH%a8%98%bd%81%5c%7d2%ee%8360
%03%09y%60%fe&peer_id=-DE13F0-
vnpZRF8ZP9iv&port=63448&uploaded=0&downloaded=0&left=105383936&corrupt=0&k
ey=C9936EC4&event=started&numwant=200&compact=1&no_peer_id=1&supportcrypt
o=1&redundant=0 HTTP/1.1
Host: files.publicdomaintorrents.com
User-Agent: Deluge 1.3.15
Accept-Encoding: gzip
Connection: close

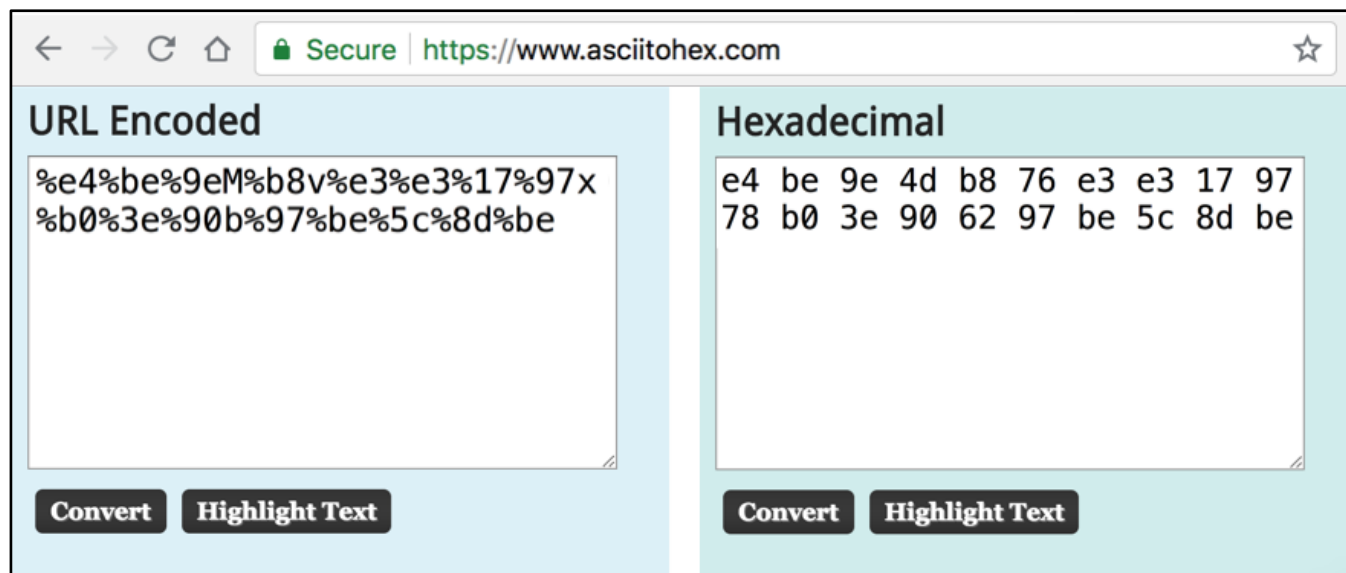
HTTP/1.1 301 Moved Permanently
Date: Sun, 15 Jul 2018 04:17:37 GMT
Server: Apache

3 client pkts, 3 server pkts, 3 turns.
```

Shown above: Selecting the SHA1 info hash from the HTTP GET request.

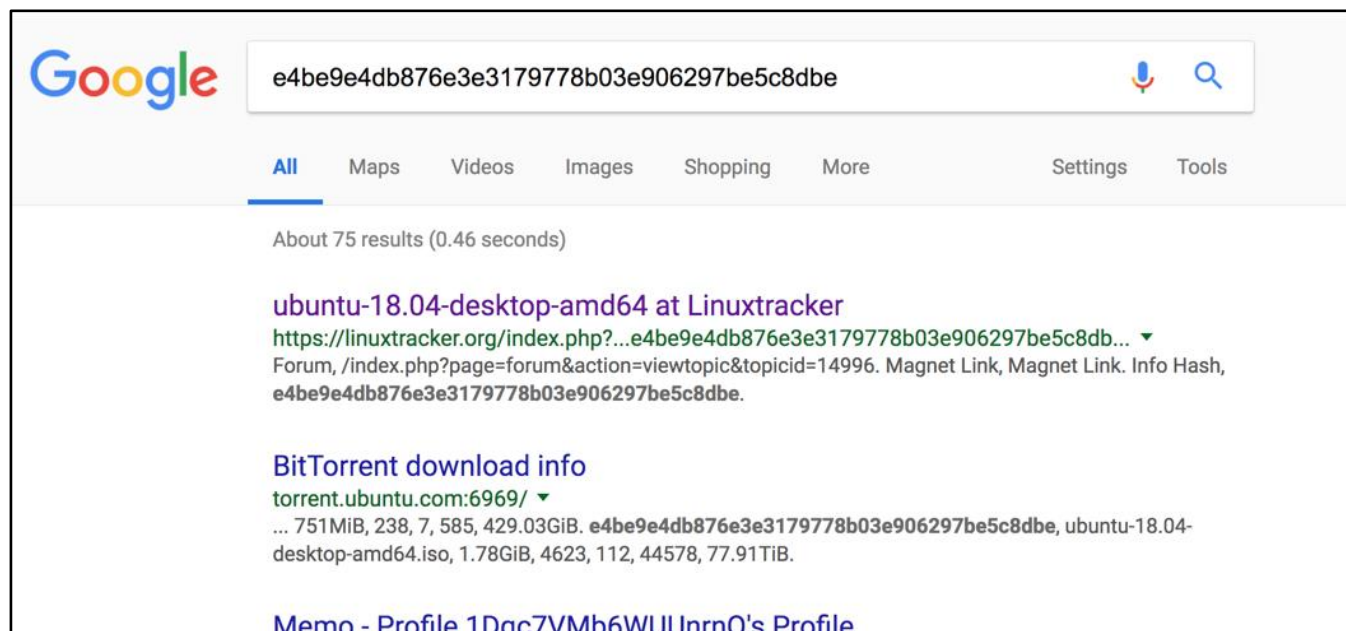
Then copy that value into an online tool like asciitohex.com to translate it to a hexadecimal value.

2018-07-15 TRAFFIC ANALYSIS EXERCISE: ANSWERS



Shown above: Translating the URL-encoded SHA1 info hash to a hex value.

The hexadecimal value is **e4be9e4db876e3e3179778b03e906297be5c8dbe** and you can search it in Google. It represents an ISO file for Ubuntu 18.04. The ISO file name is **ubuntu-18.04-desktop-and64.iso**.



Shown above: That SHA1 info hash represents **ubuntu-18.04-desktop-and64.iso**.

Do the same thing on the other HTTP requests for tracker traffic. You'll only find one other info hash: 1dda0d48a898bd815c7d32ee83366f03097960fe

That's an info hash for the Betty Boop cartoon Rhythm on the Reservation from that **.torrent** file we found earlier.

2018-07-15 TRAFFIC ANALYSIS EXERCISE: ANSWERS

http.request.uri contains scrape or http.request.uri contains announce					
Time	Dst	port	Host	Info	
2018-07-15 04:17...	91.189.95.21	6969	torrent.ubuntu.com:6969	GET	/announce?info
2018-07-15 04:17...	168.215.194.14	80	files.publicdomaintorrents...	GET	/bt/announce.p
2018-07-15 04:17...	168.215.195.227	6969	tracker.publicdomaintorrent...	GET	/announce?info
2018-07-15 04:17...	168.215.194.14	80	files.publicdomaintorrents...	GET	/bt/scrape.php
2018-07-15 04:17...	168.215.195.227	6969	tracker.publicdomaintorrent...	GET	/scrape?info_h

Shown above: Follow the TCP stream for the other torrent tracker requests.

Wireshark · Follow TCP Stream (tcp.stream eq 171) · 2018-07-15-traffic-analysis-exercise

```
GET /bt/announce.php?info_hash=%1d%da%0dH%a8%98%bd%81%5c%7d2%ee%836o
%03%09y%60%fe&peer_id=-DE13F0-
VnpZRF8ZP9iv&port=63448&uploaded=0&downloaded=0&left=05383936&corrupt=0&k
ey=C9936EC4&event=started&numwant=200&compact=1&no_ id=1&supportcrypto
=1&redundant=0 HTTP/1.1
Host: files.publicdomaintorrents.com
User-Agent: Deluge 1.3.15
Accept-Encoding: gzip
Connection: close

HTTP/1.1 301 Moved Permanently
Date: Sun, 15 Jul 2018 04:17:37 GMT
Server: Apache
```

3 client pkts, 3 server pkts, 3 turns.

Shown above: Get the info hash from the HTTP GET request.

Secure | <https://www.asciitohex.com>

URL Encoded

%1d%da%0dH%a8%98%bd%81%5c%7d2%ee%836o%03%09y%60%fe

Convert Highlight Text

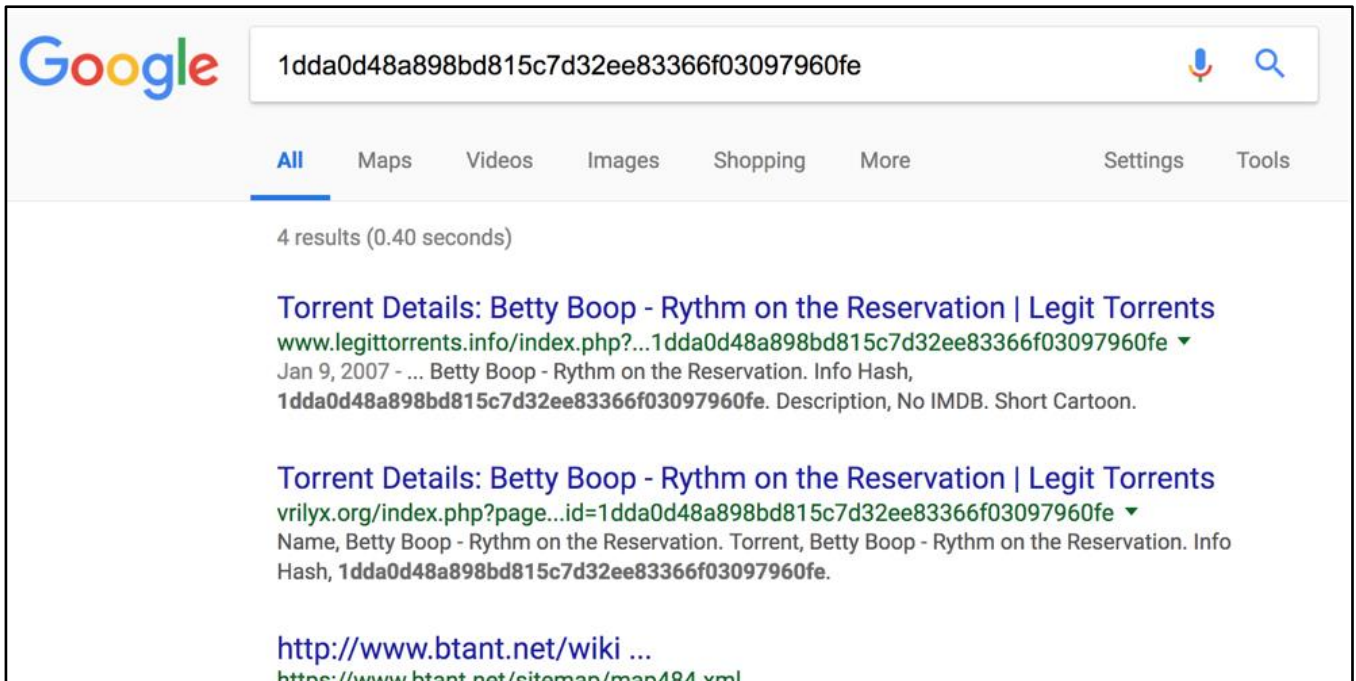
Hexadecimal

1d da 0d 48 a8 98 bd 81 5c 7d
32 ee 83 36 6f 03 09 79 60 fe

Convert Highlight Text

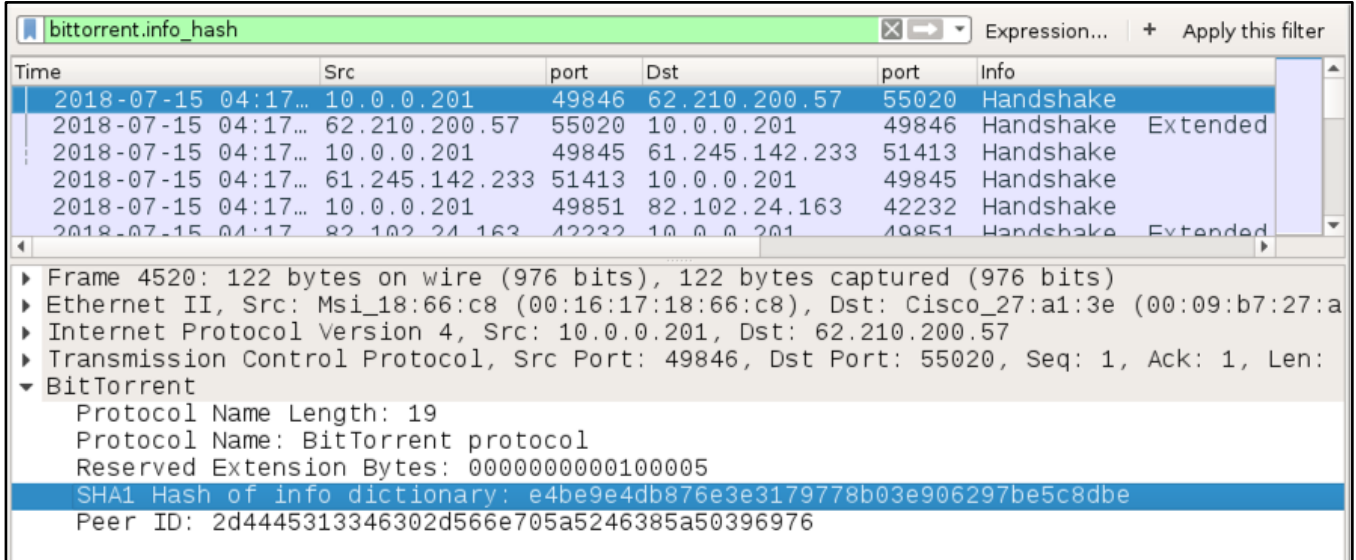
Shown above: Translate it to a hexadecimal value.

2018-07-15 TRAFFIC ANALYSIS EXERCISE: ANSWERS



Shown above: Do a Google search for the info hash to find out what it represents.

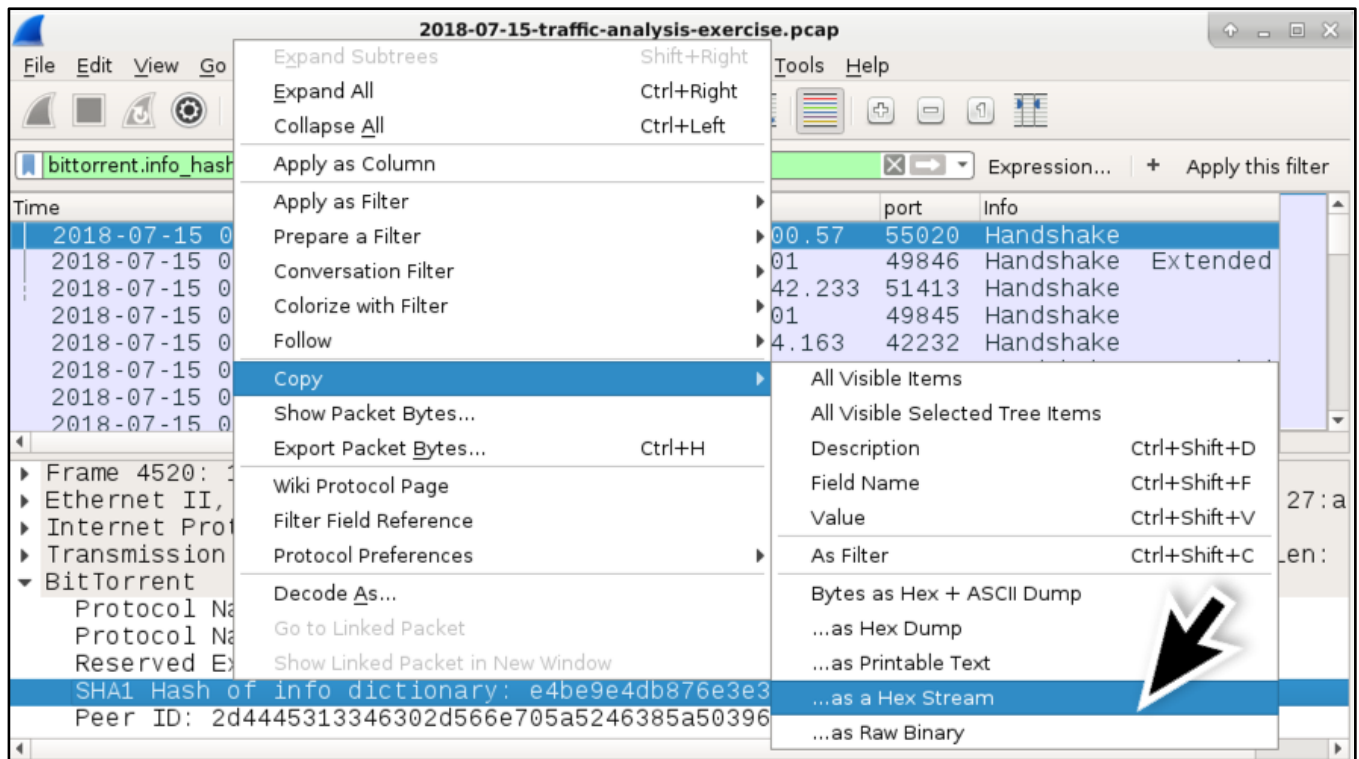
Find out if any torrent files are being seeded by filtering in Wireshark for **bittorrent.info_hash** as shown below.



Shown above: Filtering on **bittorrent.info_hash** and finding a SHA1 hash.

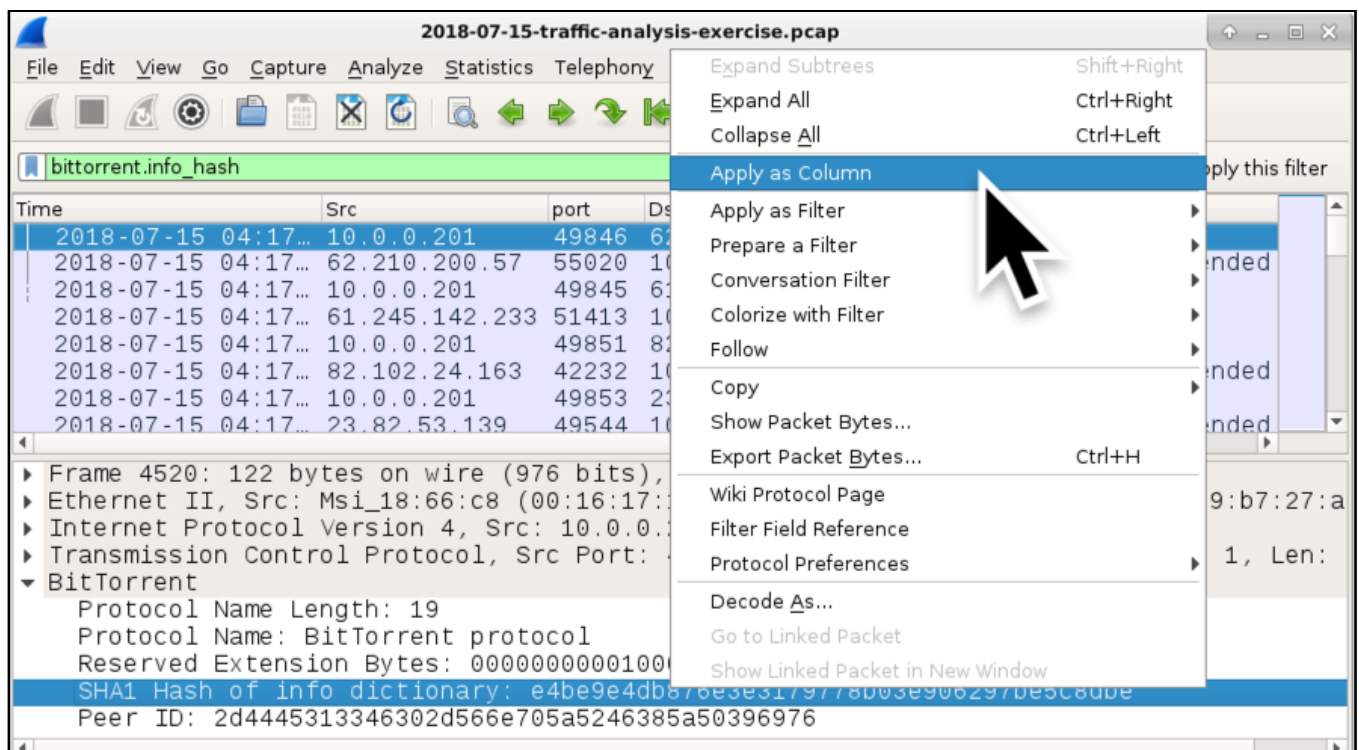
You can copy the hexadecimal value of that field by copying it as a hex stream. Then paste it into a Google search, if needed. In this case, it's the same SHA1 hash for the Ubuntu 18.04 ISO file we found earlier.

2018-07-15 TRAFFIC ANALYSIS EXERCISE: ANSWERS



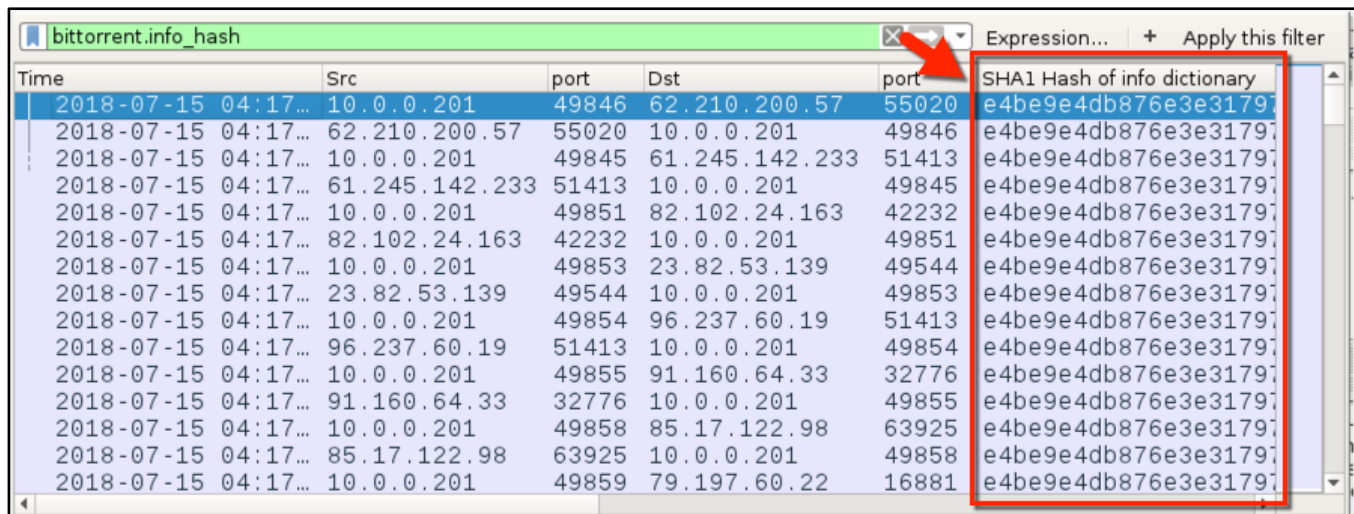
Shown above: Copy a SHA1 hash as a hex stream, so you can paste it into a Google search.

You can also apply that value as a column, so you can quickly view all of the SHA1 hashes being seeded in the pcap.



Shown above: Applying the bittorrent info hash as a column.

2018-07-15 TRAFFIC ANALYSIS EXERCISE: ANSWERS



The image shows a Wireshark packet capture interface. The filter bar at the top contains the expression 'bittorrent.info_hash'. A red arrow points to the filter bar. The packet list table has a new column 'port' (labeled 'port' in the header) which contains SHA1 hashes. The table is as follows:

Time	Src	port	Dst	port	SHA1 Hash of info dictionary
2018-07-15 04:17...	10.0.0.201	49846	62.210.200.57	55020	e4be9e4db876e3e31797
2018-07-15 04:17...	62.210.200.57	55020	10.0.0.201	49846	e4be9e4db876e3e31797
2018-07-15 04:17...	10.0.0.201	49845	61.245.142.233	51413	e4be9e4db876e3e31797
2018-07-15 04:17...	61.245.142.233	51413	10.0.0.201	49845	e4be9e4db876e3e31797
2018-07-15 04:17...	10.0.0.201	49851	82.102.24.163	42232	e4be9e4db876e3e31797
2018-07-15 04:17...	82.102.24.163	42232	10.0.0.201	49851	e4be9e4db876e3e31797
2018-07-15 04:17...	10.0.0.201	49853	23.82.53.139	49544	e4be9e4db876e3e31797
2018-07-15 04:17...	23.82.53.139	49544	10.0.0.201	49853	e4be9e4db876e3e31797
2018-07-15 04:17...	10.0.0.201	49854	96.237.60.19	51413	e4be9e4db876e3e31797
2018-07-15 04:17...	96.237.60.19	51413	10.0.0.201	49854	e4be9e4db876e3e31797
2018-07-15 04:17...	10.0.0.201	49855	91.160.64.33	32776	e4be9e4db876e3e31797
2018-07-15 04:17...	91.160.64.33	32776	10.0.0.201	49855	e4be9e4db876e3e31797
2018-07-15 04:17...	10.0.0.201	49858	85.17.122.98	63925	e4be9e4db876e3e31797
2018-07-15 04:17...	85.17.122.98	63925	10.0.0.201	49858	e4be9e4db876e3e31797
2018-07-15 04:17...	10.0.0.201	49859	79.197.60.22	16881	e4be9e4db876e3e31797

Shown above: Now we have a new column with the SHA1 info hash of the torrented files.

There may be more you could explore in this pcap; however, we've answered all the questions for this exercise.